

NORTH BAY, ON, Canada

WMO: 717310

Lat: 46.365N

Lon: 79.423W

Elev: 1215

StdP: 14.06

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-12.5	-26.5	1.3	-17.0	-21.4	1.8	-11.9	25.5	17.5	23.1	16.9	6.4	340	0.632

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	82.5	68.1	79.6	66.6	77.0	65.1	71.4	78.5	69.5	75.4	67.8	73.5	11.2	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.1	112.0	74.4	67.4	105.5	72.6	65.7	99.2	71.3	36.0	78.7	34.3	75.5	32.9	73.4	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	19.9	17.8	DB	-25.8	87.4	6.7	2.6	-30.6	89.3	-34.5	90.8	-38.2	92.3	-43.1	94.2	(4)
(5)			WB	-26.1	74.8	6.5	2.2	-30.8	76.4	-34.6	77.7	-38.3	79.0	-43.0	80.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.4	10.2	13.6	24.2	38.2	52.8	62.1	66.2	64.4	57.0	44.4	31.1	18.6		
	DBStd	21.91	13.94	11.92	12.10	9.56	8.82	7.10	5.51	5.73	7.89	8.86	10.06	11.88		
	HDD50	5299	1234	1020	803	371	74	4	0	0	25	224	569	975		
	HDD65	9227	1699	1440	1266	805	390	135	52	81	261	641	1017	1440		
	CDD50	1783	0	0	2	16	160	369	503	448	234	49	2	0		
	CDD65	235	0	0	0	0	10	50	91	63	20	1	0	0		
	CDH74	1308	0	0	0	4	83	307	505	321	85	3	0	0		
	CDH80	213	0	0	0	0	13	52	91	49	8	0	0	0		
	WSAvg	8.7	8.9	9.3	9.5	9.8	8.9	8.0	7.8	7.6	7.9	9.0	9.3	8.9		
	PrecAvg	39.8	2.6	2.2	2.5	2.7	3.3	3.7	4.0	4.0	4.5	3.7	3.6	3.1		
(15) Precipitation	PrecMax	49.5	5.9	4.3	5.0	4.9	7.1	6.3	10.1	9.2	7.5	6.1	6.6	6.3		
	PrecMin	30.3	1.0	0.5	0.5	0.8	0.7	0.9	0.6	1.0	1.9	0.6	0.6	1.7		
	PrecStd	4.8	0.9	0.9	1.2	1.0	1.5	1.5	2.1	1.7	1.6	1.3	1.3	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	44.2	59.0	71.2	81.6	84.9	86.8	85.1	80.8	72.4	58.7	46.7		
		MCWB	44.8	41.5	50.8	55.7	65.4	69.5	71.2	70.1	67.7	61.5	51.7	45.6		
	2%	DB	37.4	38.5	50.2	63.5	75.8	81.1	82.4	80.7	76.4	66.6	52.9	40.7		
		MCWB	36.5	35.5	43.8	50.0	60.1	67.1	68.4	68.1	66.4	59.1	49.1	39.3		
	5%	DB	33.0	34.0	44.7	57.5	71.7	77.6	79.2	77.6	72.5	62.1	48.7	36.9		
		MCWB	31.8	31.7	39.1	47.1	58.1	64.9	66.7	65.8	64.2	56.1	45.8	35.7		
	10%	DB	30.0	30.9	40.3	52.5	67.4	74.1	76.5	74.5	69.1	57.7	44.9	33.7		
		MCWB	28.8	29.2	35.4	43.0	56.0	62.8	65.3	64.7	62.6	53.3	42.2	32.7		
	0.4%	WB	44.6	42.4	53.1	58.7	68.3	72.9	74.3	73.1	70.9	64.2	54.0	46.4		
		MCDB	44.9	44.0	56.7	67.4	78.0	81.4	82.5	79.9	76.6	69.4	56.1	46.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.5	35.7	44.1	53.5	63.9	70.0	71.5	70.3	68.3	60.1	49.8	39.6		
		MCDB	37.2	38.1	48.3	59.2	71.5	77.5	78.4	76.5	72.6	65.3	52.3	40.2		
	5%	WB	32.0	31.9	39.7	48.4	60.9	67.5	69.5	68.4	66.0	56.9	46.1	35.9		
		MCDB	32.7	33.4	43.7	56.0	68.1	73.6	75.2	74.1	71.1	61.0	48.2	37.1		
	10%	WB	28.8	29.0	35.7	44.3	57.7	65.1	67.7	66.6	63.6	53.7	42.3	32.8		
		MCDB	30.0	31.0	39.8	51.3	65.0	71.1	73.4	71.9	68.3	57.6	44.7	33.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	16.6	16.7	17.1	18.4	17.4	16.9	16.4	16.1	13.5	12.0	12.7		
		MCDBR	14.3	16.3	18.6	23.7	23.1	20.3	19.7	19.3	18.9	19.0	17.2	12.9		
		MCWBR	14.2	14.9	13.3	15.1	12.4	10.3	9.3	9.4	10.7	12.4	14.6	12.7		
		MCDBR	14.1	15.8	17.1	21.1	18.9	17.0	16.8	15.8	16.6	17.6	16.3	12.8		
	5% WB	MCWBR	14.4	15.1	13.6	15.7	12.2	10.1	9.7	9.6	11.0	13.1	14.9	12.9		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.279	0.310	0.346	0.356	0.387	0.389	0.381	0.359	0.330	0.295	0.267		
	taud		2.349	2.292	2.323	2.369	2.413	2.348	2.370	2.387	2.434	2.504	2.527	2.418		
	Ebn at Noon		266	285	291	289	288	279	277	274	271	264	256	254		
	Edh at Noon		23	30	34	36	36	38	37	35	31	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		408	701	1159	1433	1650	1791	1785	1531	1168	697	384	303		
	RadStd		56	90	119	116	147	134	133	97	124	82	46	29		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page